

Appendix D

Rationales of the Proposed Task Conceptualisation

The proposed definition extends the conceptualisation of Related Work Generation by aligning the task with the broader literature review writing process identified in Sections 2.1 and 2.2. Rather than defining the task primarily as the automatic generation of text, AI-Assisted Literature Review Section Development conceptualises AI as supporting researchers throughout the scholarly activities required to develop a literature review section. This broader conceptualisation is grounded in the eight conceptual limitations identified in the literature.

First, the proposed definition extends the scope of the task beyond text generation by recognising that literature review section development begins before writing itself. Consequently, it explicitly incorporates literature discovery and selection, literature review planning, and critical synthesis as integral scholarly activities rather than assuming that writing begins once the relevant literature has already been identified (Limitations 1, 2, 3, and 4).

Second, the proposed definition recognises that literature review sections are scholarly arguments rather than collections of summaries. Consequently, it emphasises critical synthesis, knowledge organisation, and rhetorical decision-making while acknowledging that

multiple valid literature review sections may legitimately be produced for the same target paper depending on the author’s objectives, organisation, and interpretation of the literature (Limitations 5 and 6).

Third, the proposed definition conceptualises literature review section development as an iterative and collaborative scholarly process. Rather than viewing generation as a single-stage production task, it supports continuous refinement of both the literature review section and the underlying set of cited papers through repeated interaction between researchers and AI (Limitation 7).

Finally, the AI-in-the-Loop paradigm incorporates responsible scholarly decision-making as a fundamental characteristic of the task. Consistent with the ethical considerations identified in Section 2.2 and the AI-in-the-Loop perspective adopted in this research, AI functions as a decision-support partner that assists researchers throughout every scholarly activity while supporting informed scholarly decision-making and preserving human accountability for the final scholarly work (Limitation 8).